LABORATORY REPORT 9-14

Part 1: Frequency Modulation & Demodulation

Q1: Why does the carrier frequency change in an FM system?
In a Frequency Modulation (FM) system, the carrier frequency varies because the oscillator used in the circuit responds to the input signal. A **Voltage Controlled Oscillator (VCO)** functions as a transducer that converts an input voltage into a corresponding output frequency. When the amplitude of the message signal changes, it alters the control voltage applied to the oscillator. This variation affects the internal timing of the oscillator, causing the carrier frequency to shift or deviate from its nominal or center frequency. As a result, the carrier frequency continuously changes in accordance with the variations of the modulating signal, which is the fundamental principle of FM transmission.

Q2: What is the relationship between frequency deviation and the message signal?
The frequency deviation in an FM system is directly proportional to the amplitude of the modulating or message signal. This means that when the amplitude of the message signal increases, the carrier frequency deviates further away from its center frequency, and when the amplitude decreases, the deviation becomes smaller. However, the deviation is not dependent on the frequency of the message signal itself. Instead, the message frequency determines how rapidly the carrier frequency moves back and forth between its maximum and minimum deviation limits. In other words, amplitude controls the extent of frequency shift, while the message frequency controls the rate of change of that shift.

Q3: How does a Zero-Crossing Detector (ZCD) recover the message signal?
A Zero-Crossing Detector (ZCD) is used in FM demodulation to convert variations in frequency into variations in pulse timing. The detector generates a fixed-width pulse each time the incoming FM waveform crosses the zero-voltage level. This process produces a series of pulses where the spacing between pulses corresponds to the frequency of the input signal. When the frequency increases, the pulses occur closer together, and when the frequency decreases, the pulses become more widely spaced. The resulting pulse train has a duty cycle that reflects the instantaneous

frequency of the signal. A **Low-Pass Filter (LPF)** is then used to smooth or average these pulses, effectively reconstructing the original analog message signal.

Part 2: Sampling & Reconstruction

Q1: What is the significance of the Nyquist Rate?
The Nyquist Rate represents the minimum sampling frequency required to accurately capture and reproduce a continuous-time signal in a digital system. According to the Nyquist Sampling Theorem, the sampling frequency must be greater than twice the highest frequency component of the message signal ($f_s > 2f_m$). If the sampling frequency falls below this threshold, a phenomenon called **aliasing** occurs. Aliasing causes different frequency components to overlap, producing false or "ghost" frequencies that distort the reconstructed signal. Therefore, maintaining a sampling rate above the Nyquist limit ensures that all the information contained in the original signal can be preserved and later reconstructed without distortion.

Q2: Why is a Low-Pass Filter (LPF) necessary for signal reconstruction?
When a signal is sampled, the sampling process creates multiple high-frequency copies, also known as spectral images, of the original signal in the frequency domain. These additional frequency components can interfere with the proper reconstruction of the signal. A **Low-Pass Filter (LPF)** is therefore used during the reconstruction stage to remove these unwanted high-frequency replicas. By allowing only the baseband frequency components to pass through, the LPF effectively smooths the sampled waveform and converts the discrete-time samples back into a continuous analog signal that closely resembles the original input waveform.

Part 3: PCM Encoding & Decoding

Q1: What is Quantization Error, and how does it appear in the output signal?
Quantization error refers to the difference between the actual analog signal value and the nearest

discrete level available in the digital representation. During the quantization process, continuous amplitude values are approximated into a limited number of discrete levels. Because of this approximation, a small error is inevitably introduced. In a system using a **4-bit quantizer**, the signal can only be represented by 16 possible levels, which causes the reconstructed waveform to appear as a stepped or “staircase-like”